

k6 Load Test Report

Script: test/loadtest/health.js

Execution: 10 VUs for 30 seconds

Summary

Metric	Value
Checks Total	362
Checks Succeeded	354 (97.79%)
Checks Failed	8 (2.20%)
HTTP Requests	181
HTTP Failed	0%
Avg Response Time	665.28 ms
Min Response Time	477.71 ms
Median Response Time	552.86 ms
Max Response Time	1.05 s
P90	952.44 ms
P95	988.9 ms
Avg Iteration Duration	1.69 s

Observations

- All HTTP requests succeeded (0% failures).
- Health endpoint remained available throughout the test.
- 95% of responses completed in under 989 ms.
- 8 requests exceeded the 1000 ms threshold, causing the response-time check to fail.
- Average response time was 665.28 ms, indicating stable performance under the configured load.

k6 Dashboard Load Test Report

Script: test/loadtest/dashboard.js

Execution: Up to 150 VUs over 3 minutes

Metric	Value
Max Virtual Users	150
Duration	3 minutes
HTTP Requests	12,141
Checks Total	24,282
Checks Passed	11,145 (45.89%)
Checks Failed	13,137 (54.10%)
HTTP Fail Rate	100%
Average Response Time	485.77 ms
Minimum Response Time	261.37 ms
Median Response Time	316.81 ms
Maximum Response Time	2.52 s
P90	897.79 ms
P95	1.46 s
Average Iteration Duration	1.49 s

Threshold Results

- http_req_duration threshold FAILED (P95 = 1.46 s > 1 s)
- http_req_failed threshold FAILED (100% failed requests)
- Status is 200 check failed for all requests, indicating the dashboard endpoint likely returned unauthorized or error responses under the test configuration.

Conclusion

The dashboard load test generated significant concurrent traffic (150 VUs). Although average response times remained below one second, all requests failed the HTTP success criterion. This suggests an authentication, authorization, or endpoint configuration issue rather than a performance bottleneck. The dashboard API should be retested after verifying the JWT token and endpoint.

k6 Login Load Test Report

Script: test/loadtest/login.js

Execution: Up to 150 Virtual Users over 3 minutes

Metric	Value
Maximum Virtual Users	150
Test Duration	3 minutes
Total HTTP Requests	13,996
Total Checks	27,992
Checks Passed	13,995 (49.99%)
Checks Failed	13,997 (50.00%)
HTTP Request Failure Rate	100%
Average Response Time	288.60 ms
Minimum Response Time	256.94 ms
Median Response Time	281.63 ms
Maximum Response Time	1.11 s
90th Percentile (P90)	298.40 ms
95th Percentile (P95)	322.54 ms
Average Iteration Duration	1.29 s
Iterations	13,996

Threshold Results

- Response Time Threshold (P95 < 1000 ms): PASSED (322.54 ms).
- HTTP Failure Rate Threshold (rate < 1%): FAILED (100% failed requests).
- Status Code Check: All requests failed the 'Status is 200' validation.

Observations

- The login endpoint responded quickly under a load of 150 concurrent virtual users.
- The average response time remained below 300 ms, indicating good response performance.
- Despite acceptable response times, every request failed the HTTP success check.
- This behavior typically indicates incorrect request configuration, invalid credentials, an incorrect login endpoint, or an unexpected HTTP status code (such as 400, 401, or 404) rather than a server performance issue.

Conclusion

The login load test successfully generated sustained traffic with up to 150 concurrent virtual users. Performance metrics indicate that the server responded quickly, with a P95 response time of 322.54 ms. However, the functional validation failed because none of the requests returned the expected HTTP 200 status code, resulting in a 100% HTTP failure rate. The login request configuration and endpoint should be verified before repeating the performance test.